NAME

iapws − library for light and heavy water properties according to IAPWS.

LIBRARY

iapws library (*libiapws*, *−liapws*)

SYNOPSIS

Fortran **use iapws**

C **#include "iapws.h"**

Python **import pyiapws**

DESCRIPTION

Numerical implementation for reports:

- R2-83
  - [x] Tc in H2O and D2O
  - [x] pc in H2O and D2O
  - [x] rhoc in H2O and D2O
- G7-04
  - [x] kH
  - [x] kD
- R7-97
  - [x] Region 1
  - [] Region 2
  - [] Region 3
  - [x] Region 4
  - [] Region 5
- R11-24:
  - [x] Kw

NOTES

To use **iapws** within your fpm project, add the following to your fpm.toml file:

```
[dependencies]
iapws = { git="https://github.com/MilanSkocic/iapws.git" }
```

dp stands for double precision and it is an alias to real64 from the iso_fortran_env module.

Water phases:

l liquid

v vapor

c super critical

s saturation

n unknown

EXAMPLES

Example in Fortran:

```
program example_in_f
use iapws__common, only: dp, int32
use iapws
implicit none(type,external)

integer(int32) :: i, ngas
real(dp) :: T(1), kh_res(1), kd_res(1), wp_res(1), p(1)
real(dp) :: Ts(7), ps(7)
real(dp) :: x(3), y(3)
integer(int32) :: r(3)
character(len=1) :: s(3)
character(len=2) :: gas = "O2"
integer(int32) :: heavywater = 0
type(gas_type), pointer :: gases_list(:)
character(len=:), pointer :: gases_str

print *, ’########################## IAPWS VERSION ##########################
print *, "version ", version()

print *, ’########################## IAPWS R2-83 ##########################’
print "(a, f10.3, a)", "Tc in h2o=", Tc_H2O, " k"
print "(a, f10.3, a)", "pc in h2o=", pc_H2O, " mpa"
print "(a, f10.3, a)", "rhoc in h2o=", rhoc_H2O, " kg/m3"

print "(a, f10.3, a)", "Tc in D2O=", Tc_D2O, " k"
print "(a, f10.3, a)", "pc in D2O=", pc_D2O, " mpa"
print "(a, f10.3, a)", "rhoc in D2O=", rhoc_D2O, " kg/m3"
print *, ’’

print *, ’########################## IAPWS G7-04 ##########################’
! Compute kh and kd in H2O
T(1) = 25.0_dp + 273.15_dp
call kh(T, gas, heavywater, kh_res)
print "(A10, 1X, A10, 1X, A2, F10.1, A, 4X, A3, SP, F10.4)", "Gas=", gas, "T=

call kd(T, gas, heavywater, kd_res)
print "(A10, 1X, A10, 1X, A2, F10.1, A, 4X, A3, SP, F15.4)", "Gas=", gas, "T=

! Get and print available gases
heavywater = 0
ngas = ngases(heavywater)
gases_list => null()
gases_list => gases(heavywater)
gases_str => gases2(heavywater)
print *, "Gases in H2O: ", ngas
print *, gases_str
```

```
do i=1, ngas
print *, gases_list(i)%gas
enddo

heavywater = 1
ngas = ngases(heavywater)
gases_list => null()
gases_list => gases(heavywater)
gases_str => gases2(heavywater)
print *, "Gases in D2O: ", ngas
print *, gases_str
do i=1, ngas
print *, gases_list(i)%gas
enddo

print *, '########################## IAPWS R7-97 ##########################'
! Compute ps from Ts.
Ts(:) = [-1.0_dp, 25.0_dp, 100.0_dp, 200.0_dp, 300.0_dp, 360.0_dp, 374.0_dp]
Ts(:) = Ts(:) + 273.15_dp
call psat(Ts, ps)

do i=1, size(Ts)
print "(SP, F23.3, A3, 4X, F23.3, A3)", Ts(i), "K", ps(i), "MPa"
end do

! Compute Ts from ps
call Tsat(ps, Ts)
do i=1, size(Ts)
print "(SP, F23.3, A3, 4X, F23.3, A3)", Ts(i), "K", ps(i), "MPa"
end do

! Compute water properties at 280°C/8 Mpa
p(1) = 8.0_dp
T(1) = 273.15_dp + 280.0_dp
call wp(p, T, "v", wp_res)
print "(A5, F23.16, X, A)", "v(8MPa,280°C)=", wp_res(1)*1000.0_dp, "L/kg"

! Compute region and phase
x = [8.0_dp, 4.0_dp, 6.0_dp ]
y = [553.15_dp, 1200.0_dp, 2000.0_dp]
call wr(x, y, r)
call wph(x, y, s)
print *, r
print *, s

end program example_in_f
```

Example in C

```
#include <string.h>
#include <stdio.h>
#include "iapws.h"

int main(void){
double T = 25.0 + 273.15; /* in C*/
```

```
double p; /* p in Mpa */
char *gas = "O2";
double kh, kd, wp_res;
char **gases_list;
char *gases_str;
int ngas;
int i;
int heavywater = 0;
double x[3]= {8.0, 4.0, 6.0 };
double y[3] = {553.15, 1200.0, 2000.0};
int r[3];
char s[3];

printf("%s0, "########################## IAPWS VERSION ########################
printf("version %s0, iapws_version());

printf("%s0, "########################## IAPWS R2-83 ##########################
printf("%s %10.3f %s0, "Tc in H2O", iapws_r283_Tc_H2O, "K");
printf("%s %10.3f %s0, "pc in H2O", iapws_r283_pc_H2O, "MPa");
printf("%s %10.3f %s0, "rhoc in H2O", iapws_r283_rhoc_H2O, "kg/m3");

printf("%s %10.3f %s0, "Tc in D2O", iapws_r283_Tc_D2O, "K");
printf("%s %10.3f %s0, "pc in D2O", iapws_r283_pc_D2O, "MPa");
printf("%s %10.3f %s0, "rhoc in D2O", iapws_r283_rhoc_D2O, "kg/m3");

printf("0);


printf("%s0, "########################## IAPWS G7-04 ##########################
/* Compute kh and kd in H2O*/
iapws_g704_kh(&T, gas, heavywater, &kh, strlen(gas), 1);
printf("Gas=%sT=%fKkh=%+10.4f0, gas, T, kh);

iapws_g704_kd(&T, gas, heavywater, &kd, strlen(gas), 1);
printf("Gas=%sT=%fKkd=%+15.4f0, gas, T, kd);

/* Get and print the available gases */
ngas = iapws_g704_ngases(heavywater);
gases_list = iapws_g704_gases(heavywater);
gases_str = iapws_g704_gases2(heavywater);
printf("Gases in H2O: %d0, ngas);
printf("%s0, gases_str);
for(i=0; i<ngas; i++){
    printf("%s0, gases_list[i]);
}

heavywater = 1;
ngas = iapws_g704_ngases(heavywater);
gases_list = iapws_g704_gases(heavywater);
gases_str = iapws_g704_gases2(heavywater);
printf("Gases in D2O: %d0, ngas);
printf("%s0, gases_str);
for(i=0; i<ngas; i++){
    printf("%s0, gases_list[i]);
```

```
}

printf("%s0, "########################## IAPWS R7-97 ########################
double Ts[7] =  {-1.0, 25.0, 100.0, 200.0, 300.0, 360.0, 374.0};
double ps[7] = {1.0, 1.0, 1.0, 1.0, 1.0, 1.0, 1.0};
for(i=0; i<7; i++){
    Ts[i] = Ts[i] + 273.15;
}
iapws_r797_psat(7, Ts, ps);

for(i=0; i<7; i++){
    printf("%+23.3f %s %+23.3f %s0, Ts[i], "K", ps[i], "MPa");
}

iapws_r797_Tsat(7, ps, Ts);
for(i=0; i<7; i++){
    printf("%+23.3f %s %+23.3f %s0, Ts[i], "K", ps[i], "MPa");
}

T = 273.15 + 280.0;
p = 8.0;
iapws_r797_wp(&p, &T, "v", &wp_res, 1, 1);
printf("v(8MPa,280°C) = %+23.16f L/kg0, wp_res * 1000.0);

iapws_r797_wr(x, y, r, 3);
iapws_r797_wph(x, y, s, 3);
for(i=0; i<3; i++){
    printf("%i", r[i]);
}
printf("0);
for(i=0; i<3; i++){
    printf("%c", s[i]);
}
printf("0);

return 0;
}
```

Example in Python

```
r"""Example in python"""
import sys
sys.path.insert(0, "../py/src/")
import array
import numpy as np
import matplotlib.pyplot as plt
import pyiapws

print("########################## IAPWS VERSION ##########################")
print(pyiapws.__version__)

print("########################## IAPWS R2-83 ##########################")
print("Tc in H2O", pyiapws.Tc_H2O, "K")
print("pc in H2O", pyiapws.pc_H2O, "MPa")
print("rhoc in H2O", pyiapws.rhoc_H2O, "kg/m3")
```

```
print("Tc in D2O", pyiapws.Tc_D2O, "K")
print("pc in D2O", pyiapws.pc_D2O, "MPa")
print("rhoc in D2O", pyiapws.rhoc_D2O, "kg/m3")

print("")

print("########################## IAPWS G7-04 ##########################")
gas  = "O2"
T = array.array("d", (25.0+273.15,))

# Compute kh and kd in H2O
heavywater = False
k = pyiapws.kh(T, "O2", heavywater)
print(f"Gas={gas}T={T[0]}Kkh={k[0]:+10.4f}0)

k = pyiapws.kd(T, "O2", heavywater)
print(f"Gas={gas}T={T[0]}Kkd={k[0]:+10.4f}0)

# Get and print the available gases
heavywater = False
gases_list = pyiapws.gases(heavywater)
gases_str = pyiapws.gases2(heavywater)
ngas = pyiapws.ngases(heavywater)
print(f"Gases in H2O: {ngas:}")
print(gases_str)
for gas in gases_list:
    print(gas)

heavywater = True
gases_list = pyiapws.gases(heavywater)
gases_str = pyiapws.gases2(heavywater)
ngas = pyiapws.ngases(heavywater)
print(f"Gases in D2O: {ngas:}")
print(gases_str)
for gas in gases_list:
    print(gas)

style = {"marker":".", "ls":"", "ms":2}
T_KELVIN = 273.15
T = np.linspace(0.0, 360.0, 1000) + 273.15

solvent = {True: "D2O", False: "H2O"}

print("Generating plot for kh")
kname = "kh"
for HEAVYWATER in (False, True):
    print(solvent[HEAVYWATER])
    fig = plt.figure()
    ax = fig.add_subplot()
    ax.grid(visible=True, ls=':')
    ax.set_xlabel("T /°K")
    ax.set_ylabel("ln (kh/1GPa)")
    gases = pyiapws.gases(HEAVYWATER)
    for gas in gases:
```

```
        k = pyiapws.kh(T, gas, HEAVYWATER) / 1000.0
        ln_k = np.log(k)
        ax.plot(T, ln_k, label=gas, **style)
    ax.legend(ncol=3)
    fig.savefig(f"../media/g704-{kname:s}_{solvent[HEAVYWATER]}.png", dpi=100

print("Generating plot for kd")
kname = "kd"
for HEAVYWATER in (False, True):
    print(solvent[HEAVYWATER])
    fig = plt.figure()
    ax = fig.add_subplot()
    ax.grid(visible=True, ls=':')
    ax.set_xlabel("T /°K")
    ax.set_ylabel("ln kd")
    gases = pyiapws.gases(HEAVYWATER)
    for gas in gases:
        k = pyiapws.kd(T, gas, HEAVYWATER)
        ln_k = np.log(k)
        ax.plot(T, ln_k, label=gas, **style)
    ax.legend(ncol=3)
    fig.savefig(f"../media/g704-{kname:s}_{solvent[HEAVYWATER]}.png", dpi=100



print("########################## IAPWS R7-97 ##########################")
Ts = np.asarray([-1.0, 25.0, 100.0, 200.0, 300.0, 360.0, 374.0])
Ts = Ts + 273.15


ps = pyiapws.psat(Ts)
for i in range(Ts.size):
    print(f"{Ts[i]:23.3f} K {ps[i]:23.3f} MPa.")

Ts = pyiapws.Tsat(ps)
for i in range(Ts.size):
    print(f"{Ts[i]:23.3f} K {ps[i]:23.3f} MPa.")

fig = plt.figure()
ax = fig.add_subplot()
ax.grid(visible=True, ls=':')
ax.set_xlabel("Ts /K")
ax.set_ylabel("ps /MPa")
Ts = np.linspace(0.0, 370.0, 500)
Ts = Ts + 273.15

ps = pyiapws.psat(Ts)
ax.plot(Ts, ps, "r-", label="ps(Ts)")

Ts = pyiapws.Tsat(ps)
ax.plot(Ts, ps, "b--", label="Ts(ps)")

ax.legend()
fig.savefig(f"../media/r797-r4.png", dpi=100, format="png")
```

```
T = 280.0 + 273.15
p = 8.0
res = pyiapws.wp(p, T, "v")*1000.0
print(f"v(8MPa,280°C) = {res:+23.16f} L/kg)")

x = np.asarray([8.0, 4.0, 6.0 ])
y = np.asarray([553.15, 1200.0, 2000.0])
r=pyiapws.wr(x, y)
s=pyiapws.wph(x, y)
print(r)
print(s)


plt.show()
```

SEE ALSO

ciaaw(3), **codata**(3),